HW-1

Analysis of Peripheral Nerve Signals

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2.1 Pre-processing: use raw signals

(a) Understanding the Signals

The dataset consists of three peripheral nerve signal (PNS) recordings from a rat's sciatic nerve, acquired using a single-channel cuff electrode at a sampling rate of $f_s = 20$ kHz. Each recording contains alternating rest and stimulation periods for one of three sensory stimuli: Von Frey mechanical stimulation (VF), passive toe flexion (Flex), and nociceptive toe pinching (Pinch). These stimuli were chosen to target three different functional groups of afferent nerve fibers: Aβ tactile mechanoreceptive, Aα proprioceptive, and Aδ/C nociceptive fibers, respectively [1].

Figures 1 2 3 shows the raw time-domain signals for all three conditions with stimulus triggers overlaid. From the raw signals alone, it is difficult to visually distinguish between stimulus and rest periods due to the low signal-to-noise ratio (SNR) inherent to single-channel cuff electrode recordings, where the recorded signal is a superposition of action potentials from thousands of axons.

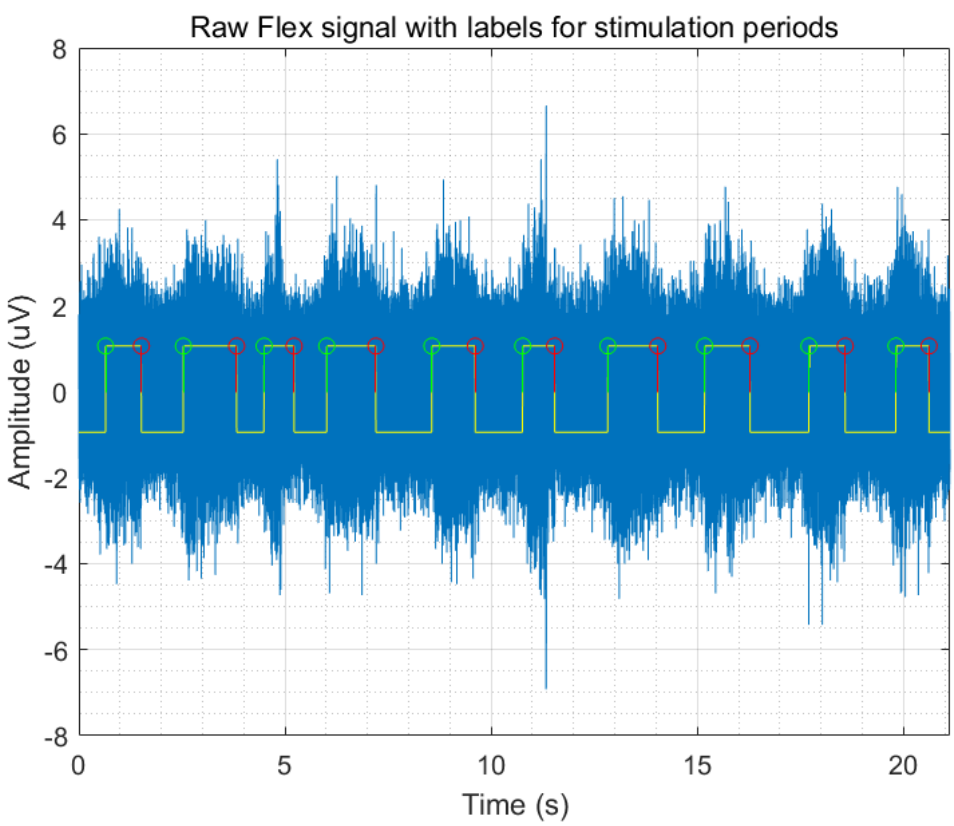

Figure 1: Raw VF signal with stimulus trigger labels. Green stems indicate stimulus onset, red stems indicate stimulus offset.

To better understand the frequency content of the signals, the Welch power spectral density (PSD) estimates were computed for the rest period and for each of the three stimulus periods (Figure 4).

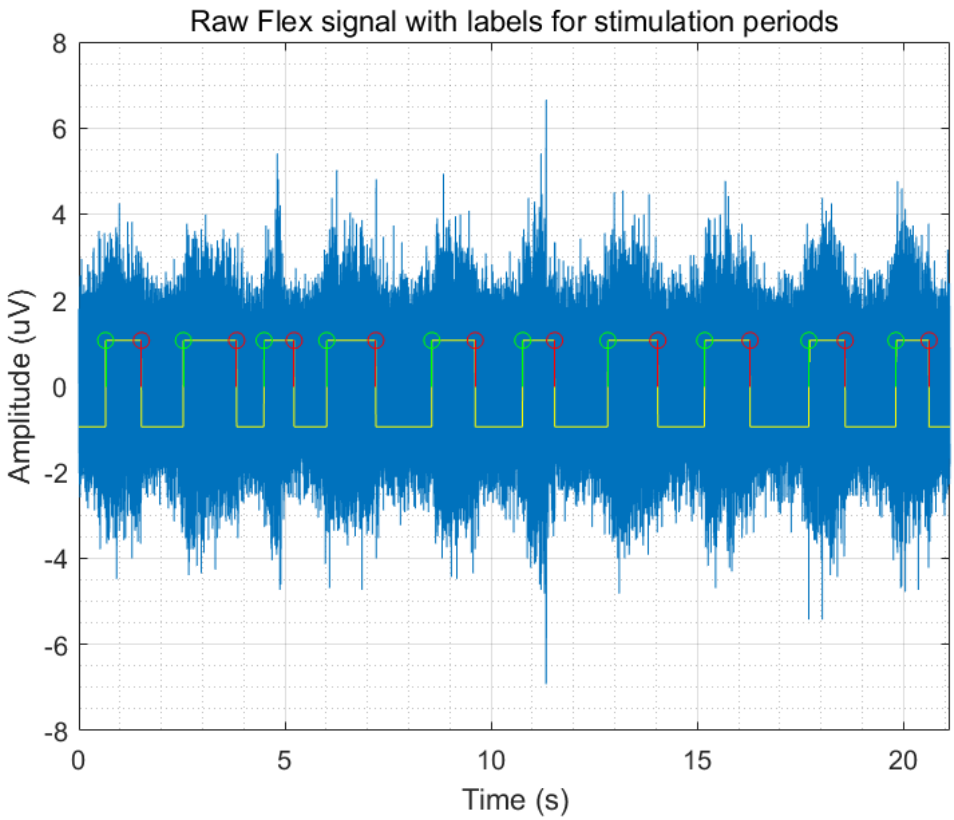

Figure 2: Raw VF signal with stimulus trigger labels. Green stems indicate stimulus onset, red stems indicate stimulus offset.

Several key observations can be made from the PSD plot. All three stimuli show higher power than the rest condition in the approximately 0.5–3 kHz range, with the peak difference occurring around 1–2 kHz. Flex and VF exhibit the most separation from rest in this frequency band, while Pinch is closest to the rest PSD, which is consistent with the thin fiber types (Aδ/C, diameter 0.3–5 μm) that carry nociceptive signals and produce smaller amplitude signals. Above approximately 4 kHz, all signals converge to a common noise floor, indicating that this higher frequency range contains mostly amplifier noise rather than useful neural information.

(b) Bandpass Filtering

Frequency band selection.
From the PSD analysis, the frequency range where stimuli and rest are most distinguishable is approximately 1–2 kHz. Observing the PSD more broadly, one might consider a wider band (e.g., 1.4–5.3 kHz) to capture more signal energy. However, following the approach in [1], a bandpass of 0.8–2.2 kHz was selected for the following reasons:

- **Below 0.8 kHz:** EMG contamination is significant, as approximately 95% of EMG power lies below 400 Hz with harmonics extending up to 800 Hz [1].

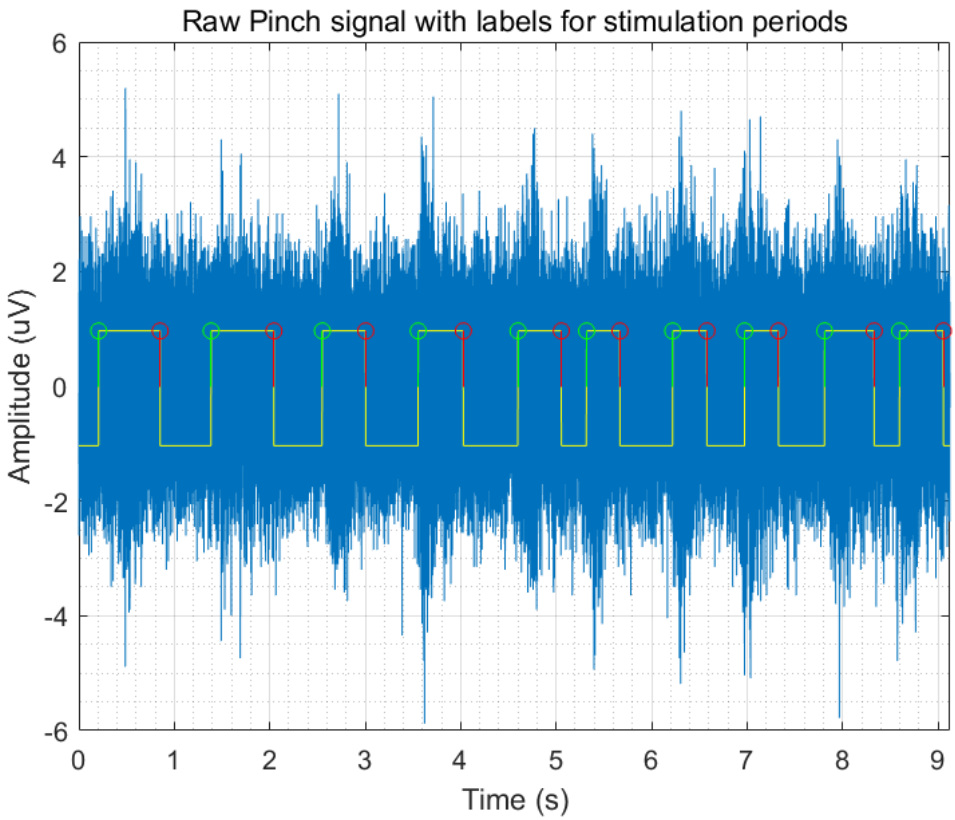

Figure 3: Raw VF signal with stimulus trigger labels. Green stems indicate stimulus onset, red stems indicate stimulus offset.

- **Above 2.2 kHz:** High-frequency amplifier noise dominates. Although some signal energy exists beyond 2.2 kHz, extending the passband would include more noise than signal, potentially degrading the SNR.

A 4th-order Butterworth bandpass filter with cutoff frequencies of 0.8 kHz and 2.2 kHz was applied using zero-phase filtering (`filtfilt`) to avoid phase distortion. **Filtering usefulness.**
Bandpass filtering removes frequency components where stimuli and rest share similar power (i.e., noise), while preserving the band where neural activity differs most between conditions. This effectively increases the SNR by eliminating energy that does not contribute to discrimination.
Observations from before/after comparison.
Figure 5 shows the VF signal in the time domain before and after filtering. The overall amplitude is reduced after filtering because the filter removes significant energy content outside the 0.8–2.2 kHz passband — particularly the low-frequency components (<0.8 kHz) that contributed substantially to the total signal power. However, even after filtering, stimulus and rest periods remain difficult to distinguish visually in the time domain due to the inherently low SNR of cuff electrode recordings. This motivates the need for feature extraction (Section 2.2), which can reveal differences that are not visible to the naked eye.

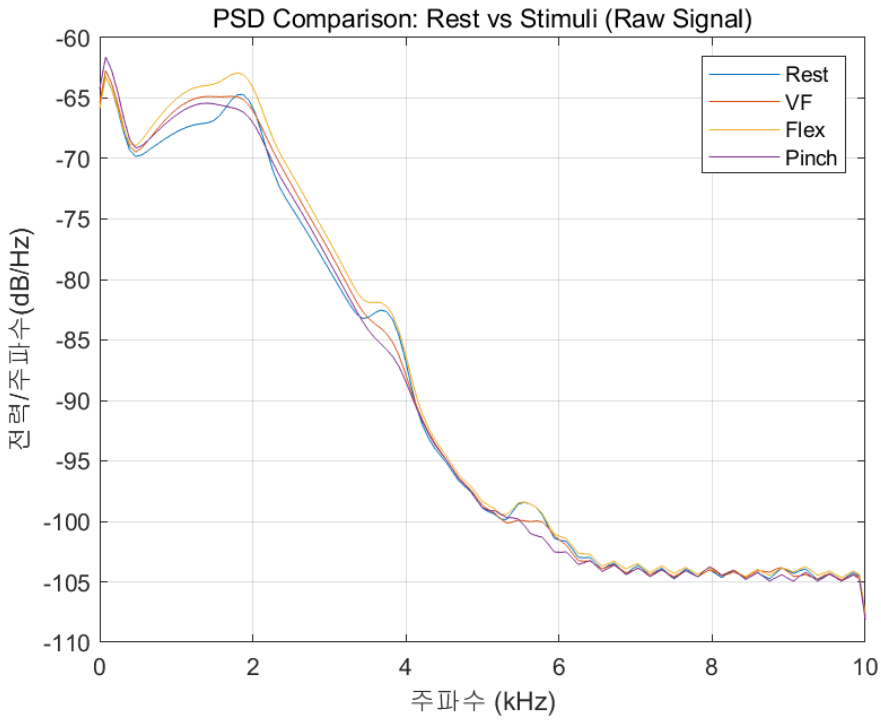

Figure 4: PSD comparison of Rest, VF, Flex, and Pinch conditions (raw signals). The stimuli show higher power than Rest in the 0.5–3 kHz range, with the peak difference around 1–2 kHz.

Figure 6 compares the PSD of the VF stimulus period before and after filtering. The filtered signal retains nearly identical power within the 0.8–2.2 kHz passband while showing a steep attenuation outside this range. A narrow spectral peak near 1.8–2 kHz is present in both the raw and filtered signals, likely originating from electromagnetic interference from the recording equipment. Since this interference falls within the passband, the bandpass filter cannot remove it; a notch filter would be needed for such narrowband noise.

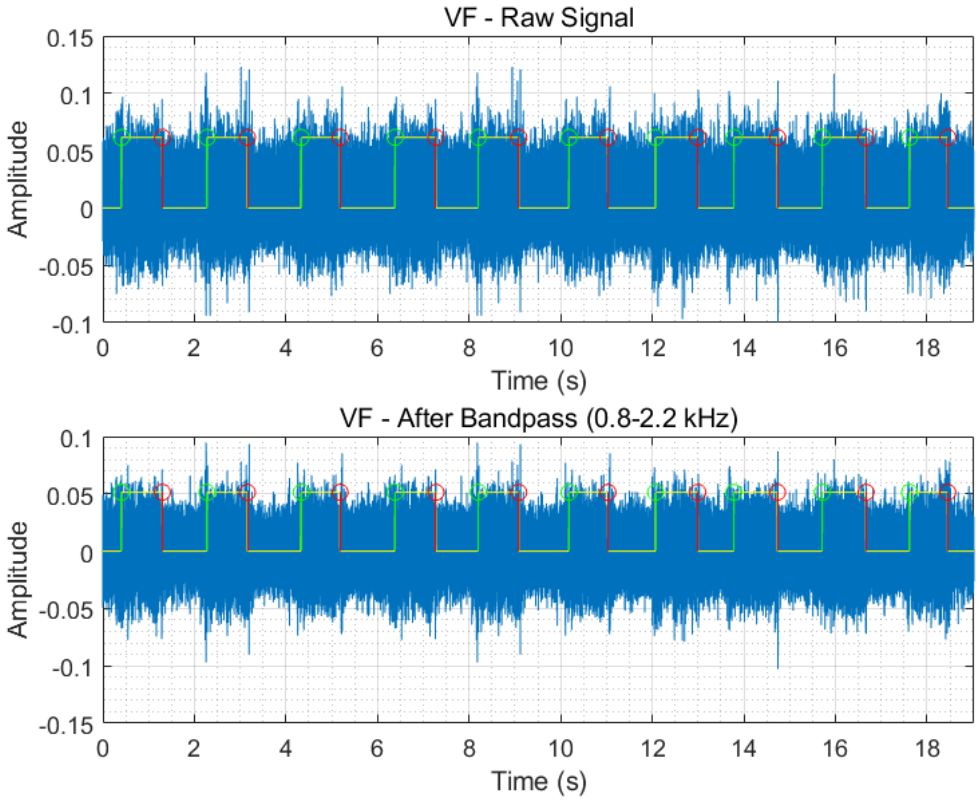

Figure 5: VF signal in the time domain before (top) and after (bottom) bandpass filtering (0.8–2.2 kHz). The overall amplitude decreases due to removal of out-of-band energy, but the relative stimulus/rest structure is preserved.

2.2 Feature Extraction: use filtered signals

(a) MAV and VAR features

Two time-domain features were computed from the bandpass-filtered signals using a sliding window approach. The Mean Absolute Value (MAV) and Variance (VAR) are defined as:

$$MAV = \frac{1}{N}\sum_{i=1}^{N}|x_i|, \qquad VAR = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 \tag{1}$$

where N is the number of samples in the window and $\bar{x}$ is the sample mean. Both features measure signal "power" but from different perspectives: MAV captures the average signal magnitude, while VAR captures the spread of the signal around its mean.

Features were computed for all three stimuli across nine combinations of window size (WSize: 50, 100, 300 ms) and overlap (Olap: 0, 0.25, 0.75). The label for each window was determined by checking whether the window was fully contained within a stimulus period.

Best WSize and Olap by visual inspection.

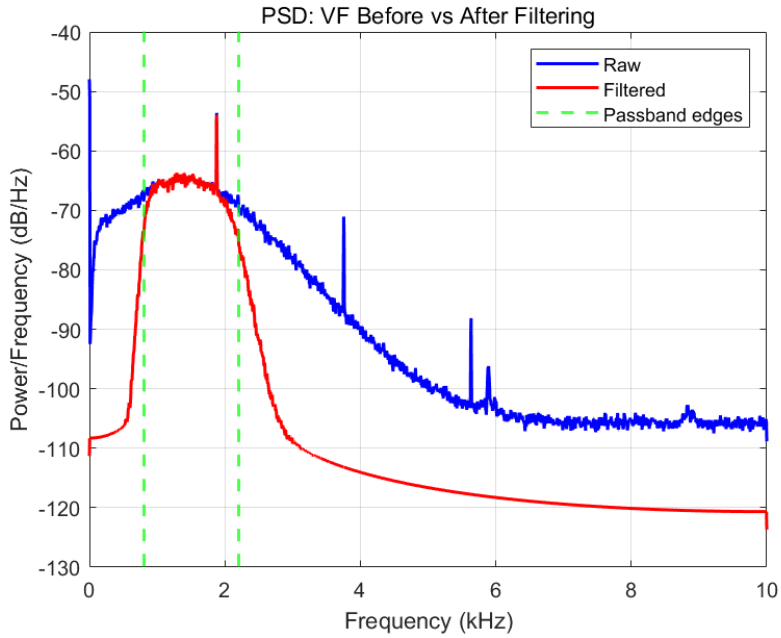

Figure 6: PSD of VF stimulus period before (blue) and after (red) bandpass filtering. Green dashed lines indicate the passband edges (0.8 and 2.2 kHz).

Based on the plots, **WSize = 100 ms and Olap = 0** provide the best balance:

- **WSize = 50 ms** produces noisy, unstable feature values with high variance, making it difficult to see consistent patterns.
- **WSize = 100 ms** provides a good trade-off between stability and temporal resolution — the feature trajectory is smooth enough to identify stimulus periods while preserving stimulus onset/offset boundaries.
- **WSize = 300 ms** is overly smooth, blurring the boundaries between stimulus and rest periods. Given that stimulus durations are approximately 1 second, a 300 ms window yields only 2–3 feature values per stimulus, losing temporal detail.
- Increasing overlap from 0 to 0.75 produces a smoother trajectory but does not substantially improve discriminability, while increasing computational cost and introducing redundancy between adjacent windows.

Better optimization approach.
Rather than relying on visual inspection, a grid search over WSize and Olap could be performed using classification accuracy as the optimization metric, similar to the approach in [1] (Figure 5). An important consideration is the real-time delay constraint: in prosthetic applications, the total delay (WSize + processing time) must remain below approximately 175–300 ms to be imperceptible to the user [1].

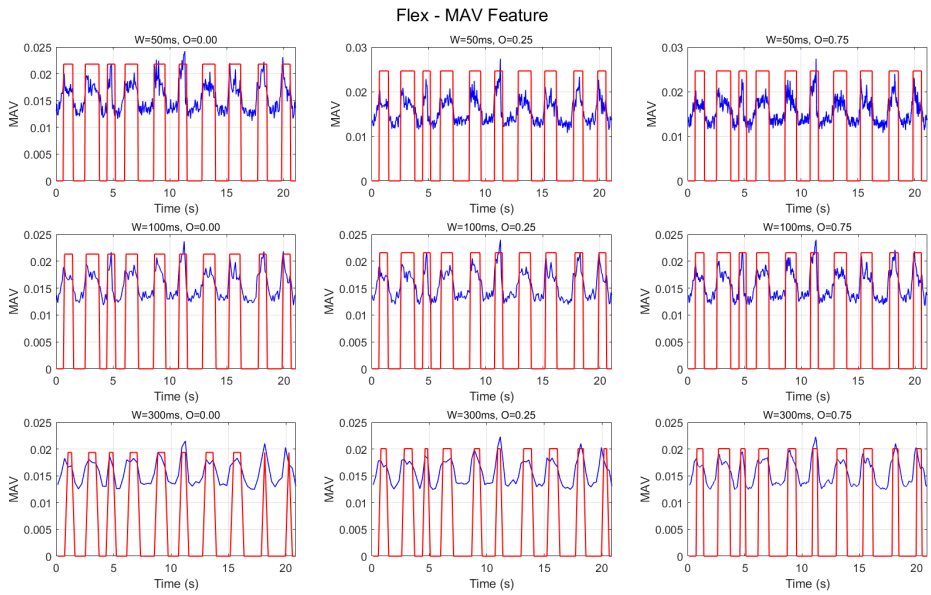

Figure 7: MAV feature for Flex across all WSize and Olap combinations. Red shading indicates stimulus periods. Stimulus periods show elevated MAV values compared to rest.

Quality of MAV and VAR for discrimination.

- **Flex (Proprioceptive):** Both MAV and VAR show clear elevation during stimulus periods compared to rest. This is the easiest stimulus to discriminate.
- **VF (Touch):** Moderate elevation during stimulus periods, though with some overlap between stimulus and rest feature values.
- **Pinch (Nociceptive):** The hardest to discriminate — stimulus and rest feature values overlap substantially, with only marginal differences visible in the plots.

This ordering is consistent with the nerve fiber diameters: proprioceptive Aα fibers (12–22 μm) produce the largest signals, followed by tactile Aβ fibers (6–12 μm), and nociceptive Aδ/C fibers (0.3–5 μm) produce the smallest [1].

(b) Feature Selection: Use WSize = 100ms and Olap = 0

The SNR metric from [1] was computed for each stimulus class:

$$SNR_{dB} = 20 \cdot \log_{10} \frac{\text{mean}(MAV_{\text{Stimulus}})}{\text{mean}(MAV_{\text{Rest}})} \tag{2}$$

The same formula was applied using VAR values in place of MAV to compare both features. The results are summarized in Table 1.

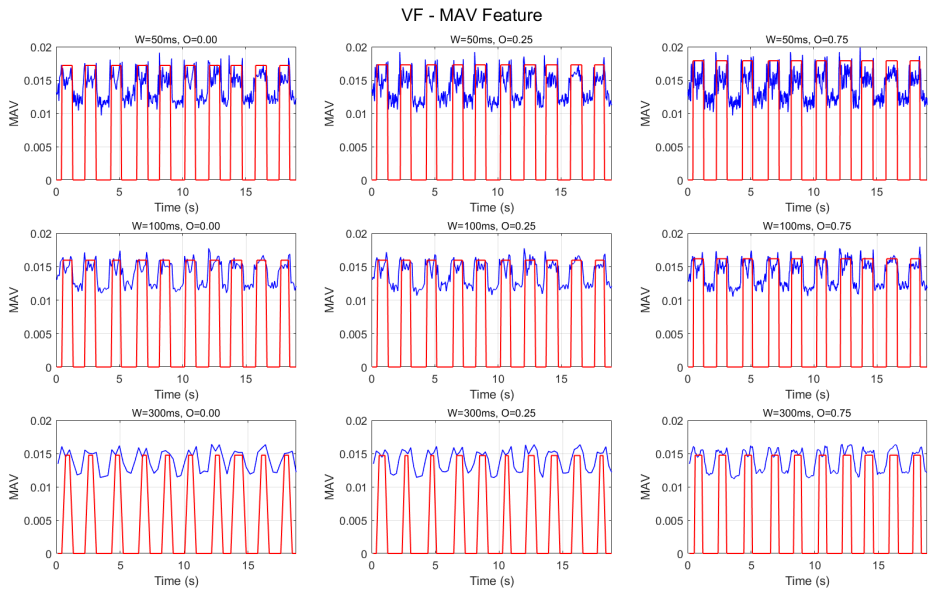

Figure 8: MAV feature for VF across all WSize and Olap combinations.

Table 1: SNR values (dB) for MAV and VAR features across the three stimulus classes.

Stimulus	$\mathbf{SNR_{MAV}}$ **(dB)**	$\mathbf{SNR_{VAR}}$ **(dB)**
VF (Touch)	1.62	3.08
Flex (Proprioceptive)	2.34	4.84
Pinch (Nociceptive)	1.12	2.48

Feature choice.

VAR yields higher SNR than MAV for all three stimulus classes. This suggests that VAR is more sensitive to the changes in neural activity caused by stimulation. While MAV measures the average signal magnitude, VAR captures the variability of the signal around its mean, which may be more responsive to the modulation of neural firing patterns during stimulation. Therefore, VAR would be the preferred feature for classification.

Easiest class to detect.

Flex (Proprioceptive) has the highest SNR for both features (MAV: 2.34 dB, VAR: 4.84 dB) and would be expected to be the easiest to classify. This has a clear physiological basis: proprioceptive signals are carried by Aα fibers, which have the largest diameter (12–22 μm) among the three fiber types tested. Larger fiber diameter is proportional to conduction velocity and signal amplitude, resulting in a stronger recorded signal relative to the background noise [1]. Conversely, Pinch has the lowest SNR (MAV: 1.12 dB, VAR: 2.48 dB) because nociceptive signals travel through thin Aδ (2–5 μm) and C fibers (0.3–1.3 μm), producing very small amplitude signals that are difficult to distinguish from background noise. The SNR values in Table 1 follow the same ranking as reported

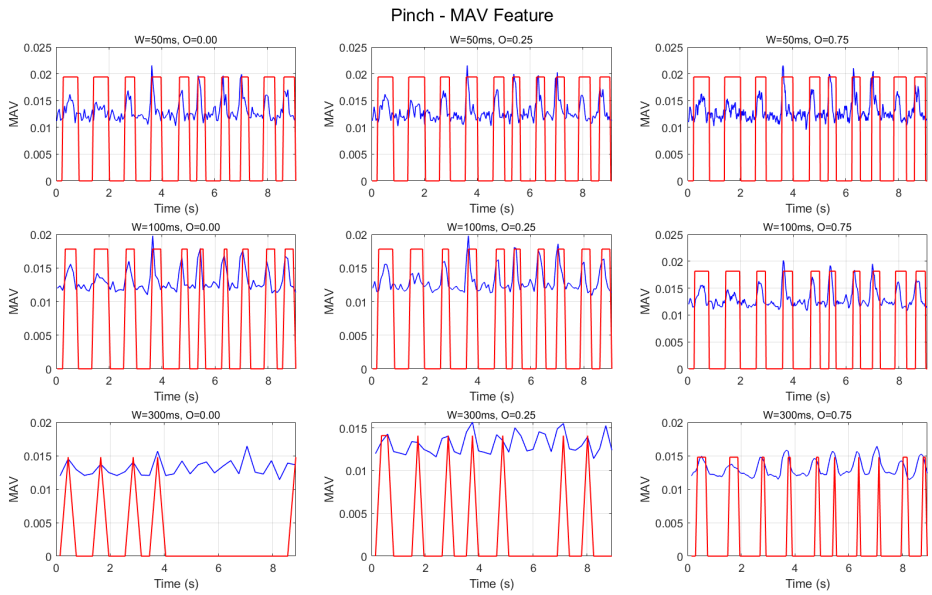

Figure 9: MAV feature for Pinch across all WSize and Olap combinations. Discrimination between stimulus and rest is notably more difficult compared to Flex and VF.

in [1] (Table 2): Proprioceptive > VF > Nociceptive.

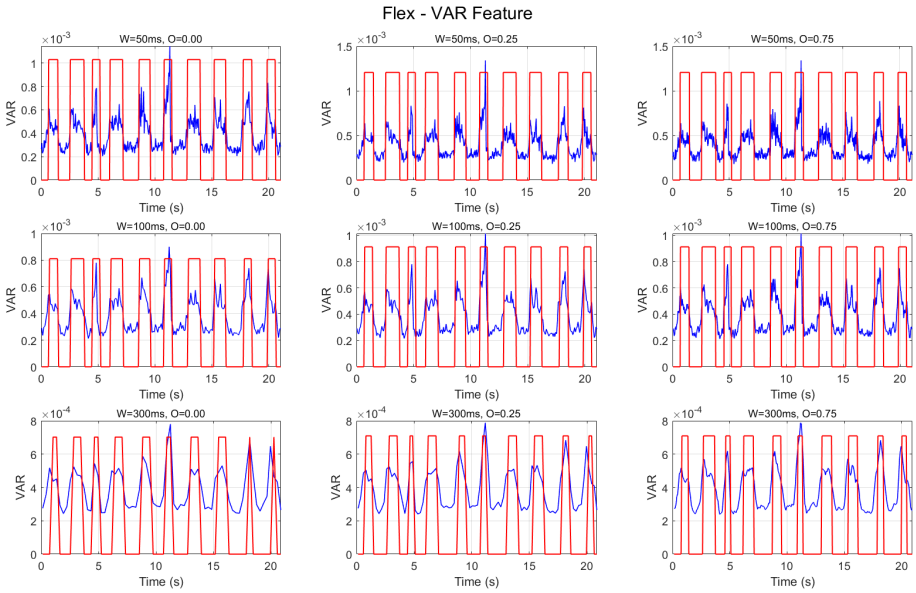

Figure 10: MAV feature for Flex across all WSize and Olap combinations. Red shading indicates stimulus periods. Stimulus periods show elevated MAV values compared to rest.

2.3 Classification: WSize = 100ms, Olap = 0

Classifiers were built using MATLAB's `classify` function (Linear Discriminant Analysis) with 10-fold cross validation. Each of the two features (MAV and VAR) was used separately to train binary classifiers for all Rest vs. Stimulus and Stimulus vs. Stimulus combinations.

Results

Rest vs. Stimulus classification:

Table 2: 10-fold cross validation accuracy (%) for Rest vs. Stimulus classification.

	MAV	**VAR**	**MAV+VAR**
VF vs Rest	82.6 ± 9.3	81.6 ± 10.3	87.4 ± 6.7
Flex vs Rest	91.5 ± 5.8	90.5 ± 7.8	91.5 ± 5.8
Pinch vs Rest	79.1 ± 11.0	78.1 ± 11.4	79.1 ± 11.0

Stimulus vs. Stimulus classification:

Can all stimuli be classified against Rest?

Yes. All three stimuli achieve classification accuracy well above the 50% chance level: Flex at 91.5%, VF at 82.6%, and Pinch at 79.1%. However, the performance varies considerably, with Pinch being the most challenging to detect.

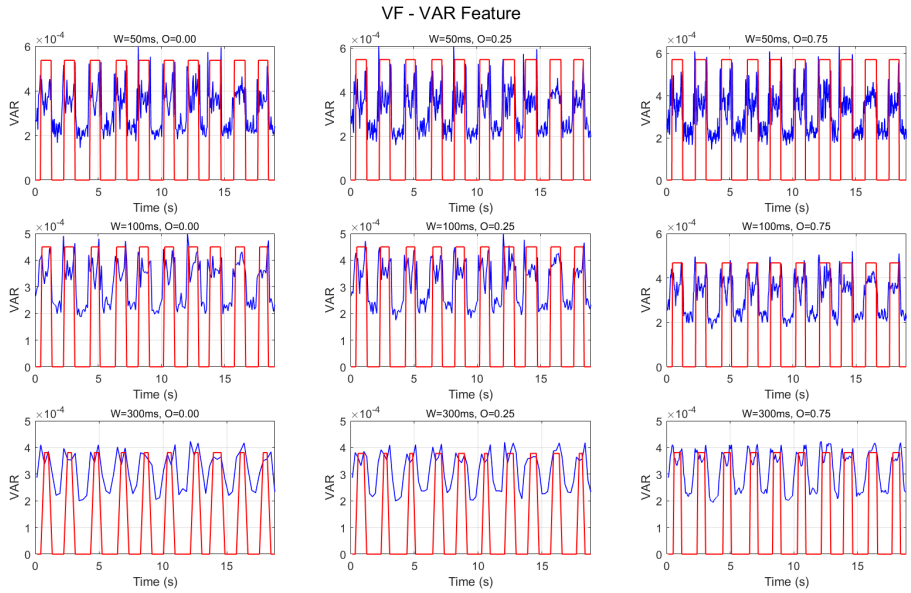

Figure 11: MAV feature for VF across all WSize and Olap combinations.

Table 3: 10-fold cross validation accuracy (%) for Stimulus vs. Stimulus classification.

	MAV	**VAR**
VF vs Flex	83.9 ± 10.7	85.7 ± 10.0
VF vs Pinch	74.3 ± 7.6	70.0 ± 9.0
Flex vs Pinch	85.9 ± 10.7	86.7 ± 9.6

Can stimuli be classified from one another?

Yes, but with varying difficulty. Flex vs Pinch achieves the highest accuracy (85.9–86.7%) because these two stimuli are mediated by fiber types with the greatest difference in diameter (Aα vs Aδ/C). VF vs Pinch has the lowest accuracy (70.0–74.3%), as both produce relatively low-amplitude signals compared to Flex.

Balance of classification rates.

The confusion matrices reveal imbalanced classification rates. For example, in Pinch vs Rest (Figure 16), the classifier correctly identifies most Rest windows but misclassifies a substantial proportion of Pinch stimulus windows as Rest (14 out of 37, ≈38% miss rate). This occurs because Pinch's low SNR causes its feature values to overlap heavily with Rest, making the classifier biased toward the Rest class.

Easiest stimulus to detect.

Flex is the easiest to detect, achieving the highest accuracy in both Rest vs. Stimulus (91.5%)

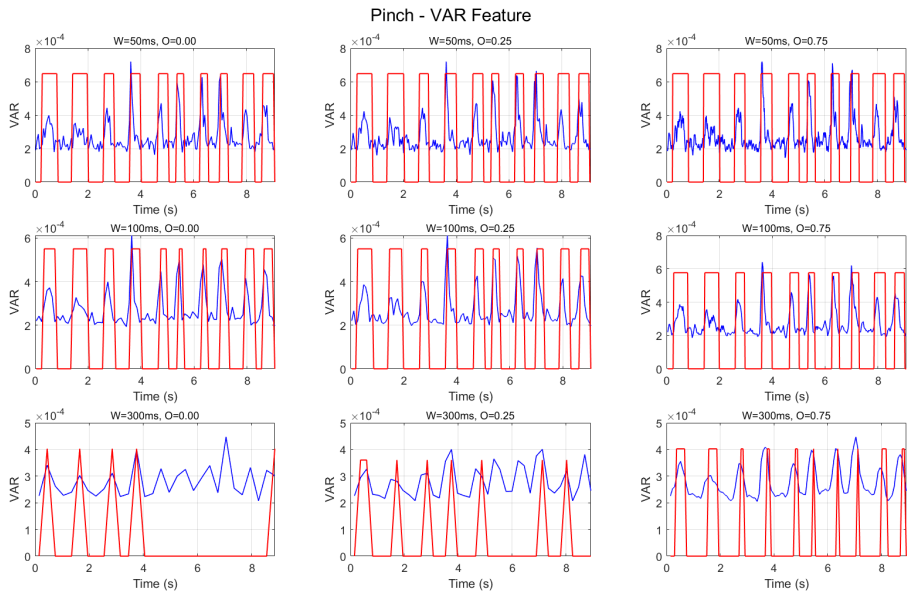

Figure 12: MAV feature for Pinch across all WSize and Olap combinations. Discrimination between stimulus and rest is notably more difficult compared to Flex and VF.

and Stimulus vs. Stimulus comparisons. This is fully consistent with the PSD plots (Section 2.1a), where Flex showed the greatest power difference from Rest in the 1–2 kHz band, and with the SNR values (Section 2.2b), where Flex had the highest SNR (MAV: 2.34 dB, VAR: 4.84 dB).

Which feature resulted in higher accuracy?

The two features perform comparably. For Rest vs Stimulus, MAV is slightly better, while for Stimulus vs Stimulus, VAR is marginally better. This is somewhat unexpected given that VAR had consistently higher SNR in Section 2.2b. The explanation is that the `classify` function uses Linear Discriminant Analysis, which can learn an effective decision boundary even with lower SNR, partially compensating for the difference. Notably, combining both features (MAV+VAR) improved VF vs Rest accuracy from 82.6% to 87.4%, suggesting that the two features carry partially complementary information.

Fairness of k-fold cross validation.

The `cvpartition` function splits the data **randomly** into training and testing sets. This is problematic for time-series data because adjacent windows are temporally correlated — they share overlapping or nearly identical underlying signal segments. With random splitting, temporally adjacent (and therefore highly similar) windows can end up simultaneously in the training and test sets, causing the classifier to effectively "memorize" test data. This leads to an **overly optimistic** estimate of the classifier's generalization ability. A fairer assessment would use time-structured splits, such as training on the first half and testing on the second half (as done in [1]), or leave-one-

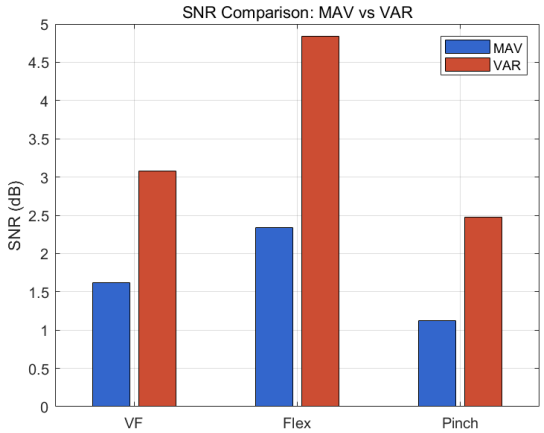

Figure 13: SNR comparison between MAV and VAR features. VAR consistently yields higher SNR across all stimulus classes.

trial-out cross validation, where entire stimulus trials are held out for testing. These approaches ensure that the test data is truly independent of the training data and provide a more realistic estimate of how the classifier would perform on future, unseen recordings.

Bonus: 4-Class Classification

Two single-feature classifiers and one combined-feature classifier were built to discriminate all four classes simultaneously (Rest, VF, Flex, Pinch). The rest periods from all three recording conditions were pooled together to form the Rest class.

Table 4: 10-fold cross validation accuracy (%) for 4-class classification.

Feature	**Accuracy (%)**
MAV	67.1 ± 9.1
VAR	68.9 ± 8.7
MAV+VAR	68.5 ± 6.7

The 4-class accuracy (67–69%) is substantially lower than the binary classification results, which is expected as the number of classes increases. The confusion matrix (Figure 21) reveals that Flex (Class 3) is classified with the highest per-class accuracy (76.7%), consistent with all prior analyses. The most significant source of error is Pinch (Class 4): only 7 out of 37 Pinch windows (18.9%) are correctly identified, with 18 misclassified as Rest. This confirms that Pinch's feature values are largely indistinguishable from Rest due to the low SNR of thin nociceptive fibers.

Does combining MAV+VAR improve performance?

No. The combined classifier (68.5%) does not improve over single-feature classifiers (67.1% and 68.9%). This is because MAV and VAR both measure aspects of signal power and therefore carry

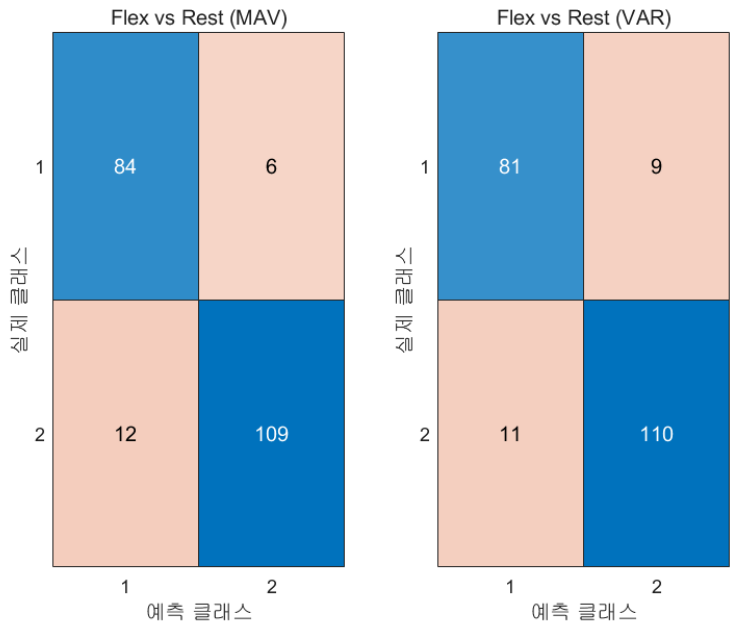

Figure 14: Confusion matrices for Flex vs Rest classification using MAV (left) and VAR (right). Class 1 = Stimulus, Class 2 = Rest.

highly correlated information. Adding a redundant feature does not provide new discriminative information; it only increases the dimensionality without meaningful benefit. To improve 4-class performance, features capturing fundamentally different signal characteristics (e.g., frequency-domain features, wavelet coefficients, or autocorrelation-based features as explored in [1]) would be needed.

References

[1] S. Raspopovic, J. Carpaneto, E. Udina, X. Navarro, and S. Micera, "On the identification of sensory information from mixed nerves by using single-channel cuff electrodes," *Journal of NeuroEngineering and Rehabilitation*, vol. 7, no. 1, p. 17, 2010.

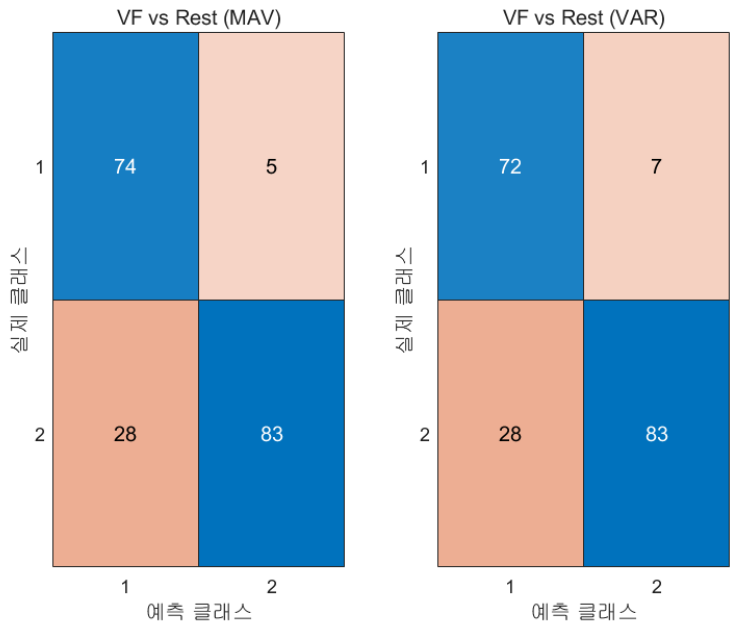

Figure 15: Confusion matrices for VF vs Rest classification using MAV (left) and VAR (right).

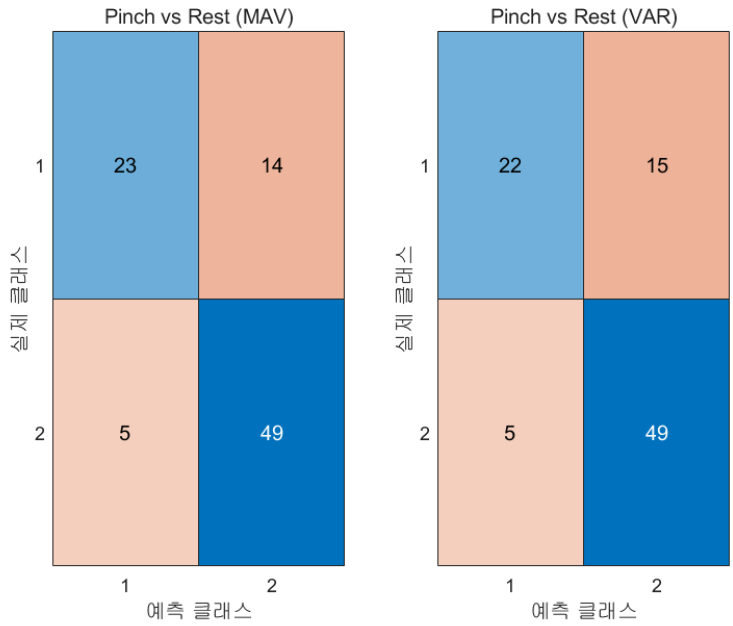

Figure 16: Confusion matrices for Pinch vs Rest classification using MAV (left) and VAR (right).

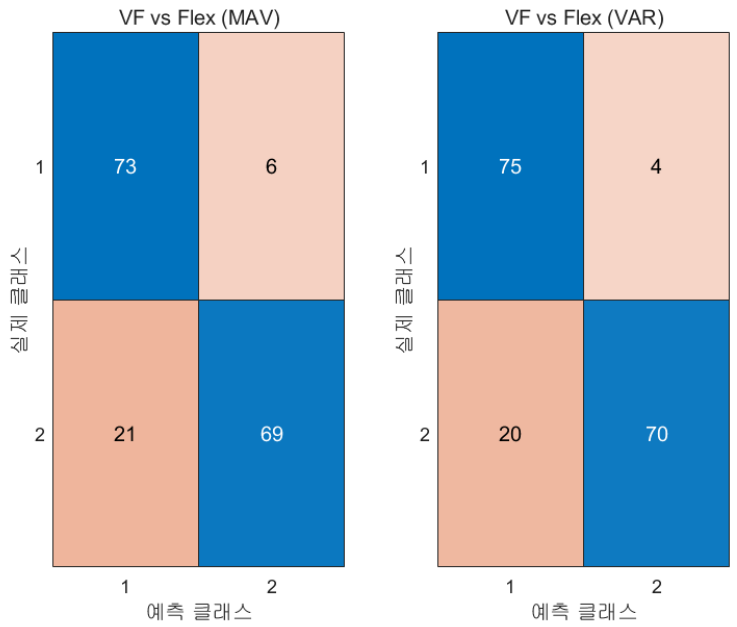

Figure 17: Confusion matrices for VF vs Flex classification.

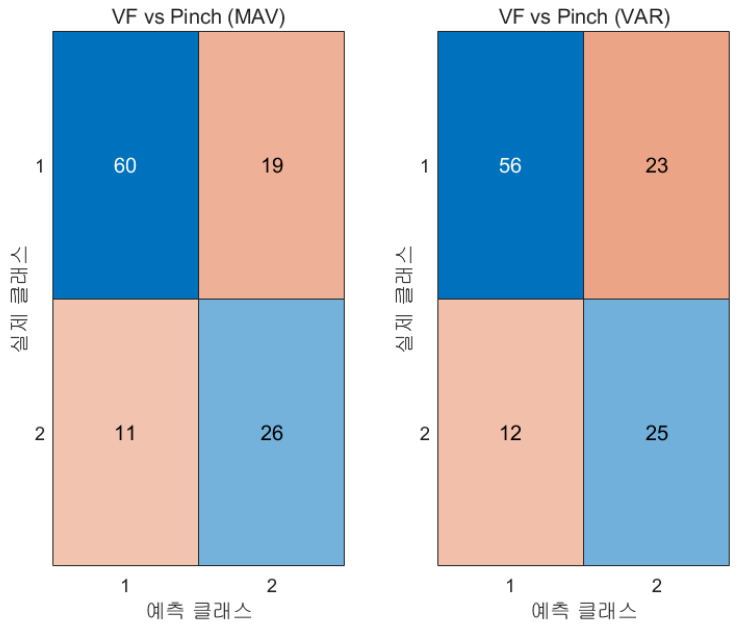

Figure 18: Confusion matrices for VF vs Pinch classification.

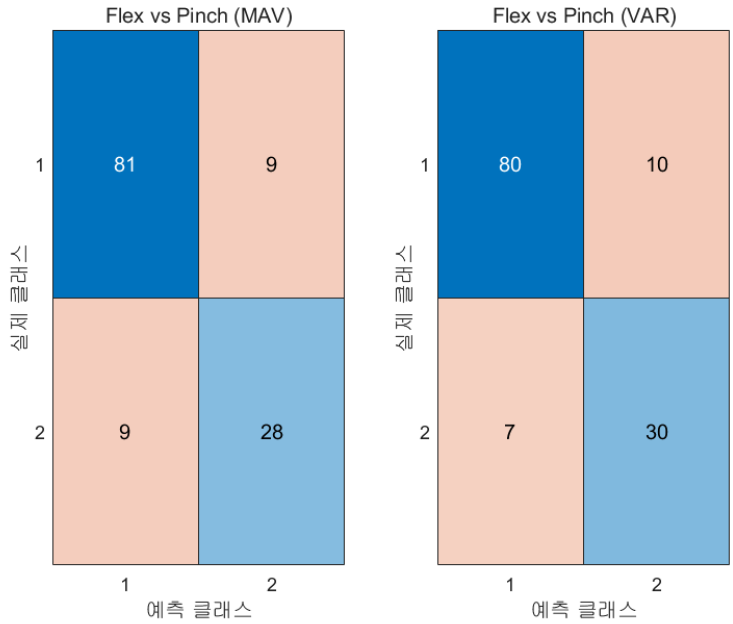

Figure 19: Confusion matrices for Flex vs Pinch classification.

Figure 20: Confusion matrix for 4-class classification using VAR. Class labels: 1 = Rest, 2 = VF, 3 = Flex, 4 = Pinch. Row-normalized percentages are shown on the right.

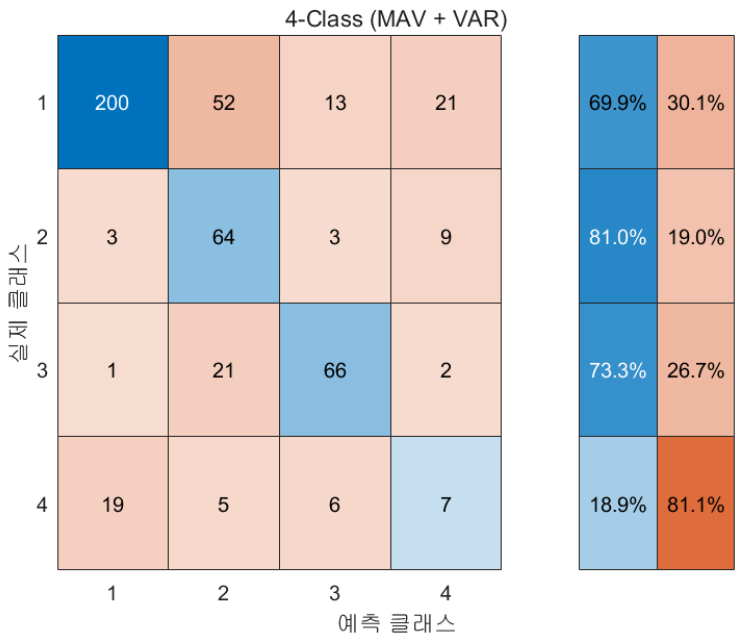

Figure 21: Confusion matrix for 4-class classification using VAR. Class labels: 1 = Rest, 2 = VF, 3 = Flex, 4 = Pinch. Row-normalized percentages are shown on the right.